N M EMRAN HUSSAIN

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WORK STATUS: Work permit (EAD) Issued | Available Immediately | Relocation available.

SUMMARY

Data Scientist & MSBA graduate specializing in Machine Learning, Generative AI (RAG), Churn Optimization, and Customer Analytics, with a proven track record of translating complex ML pipelines into measurable revenue growth.

EDUCATION

THE GEORGE WASHINGTON UNIVERSITY, School of Business — **Washington, DC**
Master of Science, Business Analytics — *December 2025*
Relevant Courses: Responsible Machine Learning, Practical Artificial Intelligence, Working with Large Dataset (Big Data), AI in Marketing, Customer Analytics & Text Analytics (NLP).

NORTH SOUTH UNIVERSITY, School of Business — **Dhaka, Bangladesh**
Master of Business Administration — *May 2023*
Relevant Courses: Marketing Research, Brand Management, Global Marketing, Marketing Strategy, Marketing Analytics.

AMERICAN INTERNATIONAL UNIVERSITY - BANGLADESH, School of Engineering — **Dhaka, Bangladesh**
Bachelor of Science, Electrical and Electronics — *March 2009*

TECHNICAL SKILLS

Programming & Code Libraries: Python (Pandas, NumPy, Scikit-learn, Matplotlib, TensorFlow, Keras), R (tidyverse, ggplot2)
Machine Learning & AI: Generative AI & LLMs (RAG, Transformers), Deep Learning (TensorFlow/Keras, PyTorch), NLP (Word2Vec, FastText, Sentiment Analysis), Responsible AI (Fairness Metrics, SHAP).
Tools & Platforms: Spark (PySpark), PostgreSQL, ETL Pipelines, Version Control (Git), GitHub, Google Cloud Platform (GCP).
Business Strategy Analytics: A/B Testing, Customer Lifetime Value (CLV), Customer Acquisition Cost (CAC), AARRR Funnel, Churn Prediction, ROMI, ROI Analysis, Stakeholder Management.

ACADEMIC PROJECTS

Predictive-RAG Retention Engine | Gemini 2.5 Flash, Reinforcement Learning, PySpark, ChromaDB, LangChain, GCP
- Engineered AI retention stack (LinUCB/Classification) that cut churn by 34%, driving $2M in incremental revenue.
- Automated e-commerce retention and decision support via LinUCB Bandits and Gemini-based RAG.

NLP Sentiment Analysis Optimization | FastText, Python, PySpark, Autotune
- Engineered a sentiment pipeline boosting IMDb accuracy from 50% to 87.4% via optimized supervised learning.
- Standardized 25,000 text samples using FastText Autotune and N-grams to eliminate underfitting and normalize data.

Loan Default Prediction Model | H2O.ai, Python & Responsible AI
- Compared H2O.ai GLM/ANN models for loan defaults, optimizing a 0.15 threshold to slash financial risk.
- Engineered an L1/L2 pipeline for feature selection and used SHAP to audit model decisions for algorithmic fairness.

Handwritten Digit Classifier | TensorFlow, Keras & Computer Vision
- Architected a TensorFlow LeNet-5 CNN for MNIST digits, achieving 96.2% accuracy with high validation generalization.
- Optimized a multi-layer pipeline using sigmoid/tanh & MaxPooling to extract spatial features & mitigate vanishing gradients.

Fairness-Aware Mortgage Pricing Model | Explainable Boosting Machines (EBM) & XAI
- Engineered an EBM for high-priced mortgage loans, balancing a 0.78 AUC with federal Fair Lending standards.
- Mitigated bias via AIR analysis and fairness remediation while using extraction attacks to verify security.

D.C. Bikeshare Optimization & Forecasting | Python & Scikit-Learn
- Forecasted urban mobility demand using ensemble models (GBM, SVM, RF) optimized for cost reduction.
- Optimized bike/dock station balance via K-medoids clustering and multi-model classification.

PROFESSIONAL EXPERIENCE

COMMUNITY DREAMS FOUNDATION | Machine Learning Engineer | July 2026 - Present
- Set up Docker containerization and initial GCP Vertex AI pipeline templates to streamline upcoming model deployments.
- Optimized exploratory feature extraction across 500K+ records in BigQuery (SQL/Python), cutting query latency by 30%.
- Configured Vertex AI Model Registry and baseline evaluation workflows to standardize tracking and validation protocols.

AMPERE LTD. | Founder & CEO (Business Analyst) | Dhaka, Bangladesh | January 2019 – December 2019
- Optimized sales trends and freight simulations using Python & R, boosting fulfillment to 90% & cutting overstocking by 5%.
- Automated cost and quality tracking via Python/R, securing 10% margins and accelerating customs clearance.

ANRHUS & SHUZOS CO. | Assistant Manager | Dhaka, Bangladesh | June 2009 – December 2018
- Built Python/R/SQL ETL pipelines, dashboards (Streamlit, Plotly), and track budgets for a $500K project.
- Built **Python/R** inventory models, boosting efficiency by up to 8%, cutting costs by 5%, and ensuring trade compliance.

CERTIFICATION

Google Professional Machine Learning Engineering (on-going)